Final Exam 2026: Part 2

GFS189, RTL959

June 24, 2026

1 Exercise 1: Volatility Signature Plot and Sampling Choice

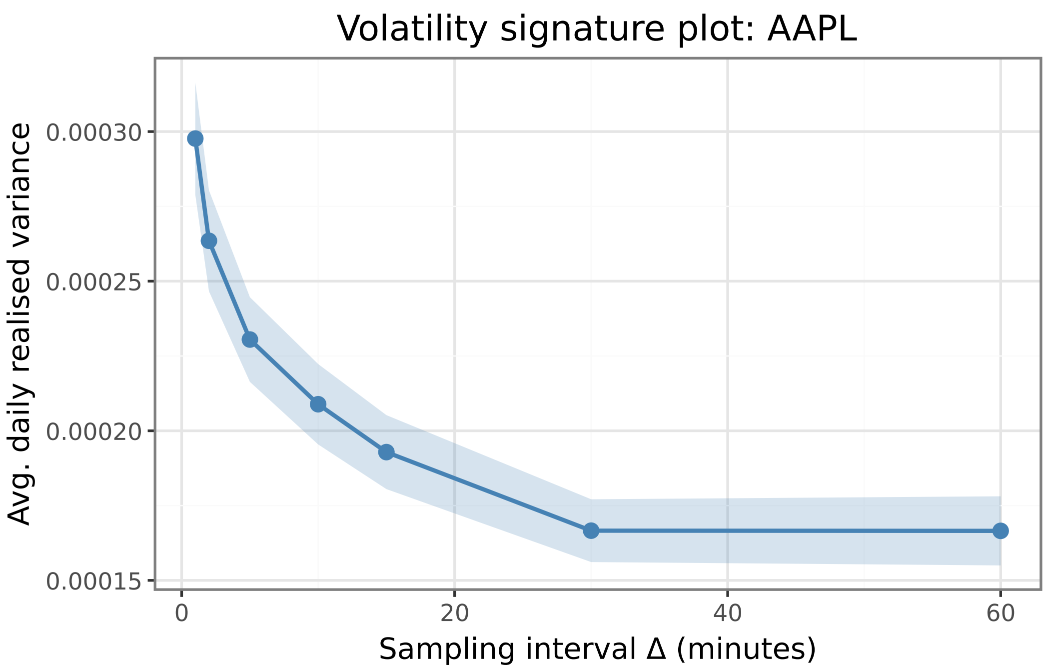

Figure 1: Volatility signature plot for AAPL. Average daily RV per interval Δ; band $= \pm 2$ SE. Upward bias at $\Delta = 1$ min reflects microstructure noise; series stabilises by $\Delta = 5$ min.

The signature plot (Figure 1) displays the characteristic bias of the **realized variance (RV)** estimator: at $\Delta = 1$ minute, bid-ask bounce inflates the sum of squared log-returns above the true integrated variance (Andersen and Bollerslev, 1998). The curve stabilises by $\Delta = 5$ minutes, motivating our choice of $\Delta = 5$ **minutes** throughout; missing minutes are handled by **previous-tick interpolation** (carry forward last observed price).

More noise-robust alternatives exist: the two-scale realized variance (TSRV) of Zhang, Mykland, and Ait-Sahalia (2005) corrects the bias by subsampling at multiple frequencies, while the multivariate realized kernel of Barndorff-Nielsen et al. (2011) additionally handles asynchronous trading via refresh-time sampling, yielding a consistent positive semi-definite estimator even at high frequencies. We adopt simple 5-minute sparse sampling instead because (i) the signature plot confirms the noise bias is negligible at this frequency for DJIA stocks, (ii) the sum-of-outer-products estimator is computationally straightforward for a 30-stock universe, and (iii) the data do not provide tick-level timestamps needed to exploit refresh-time synchronisation efficiently.

2 Exercise 2: Daily Realized Covariance

The **realized covariance (RC)** estimator $RC_d = \sum_{k=1}^{K} r_k r_k'$ sums outer products of synchronised 5-minute log-return vectors (Andersen and Bollerslev, 1998; Barndorff-Nielsen et al., 2011). Gaps within each trading day are filled by previous-tick interpolation (forward-fill only), so no backward look into future prices is introduced. Any residual missing return that remains after forward-filling — occurring when a stock has no recorded trade before the first 5-minute bar of the day — is replaced with zero, a conservative choice that does not artificially inflate RC_d.

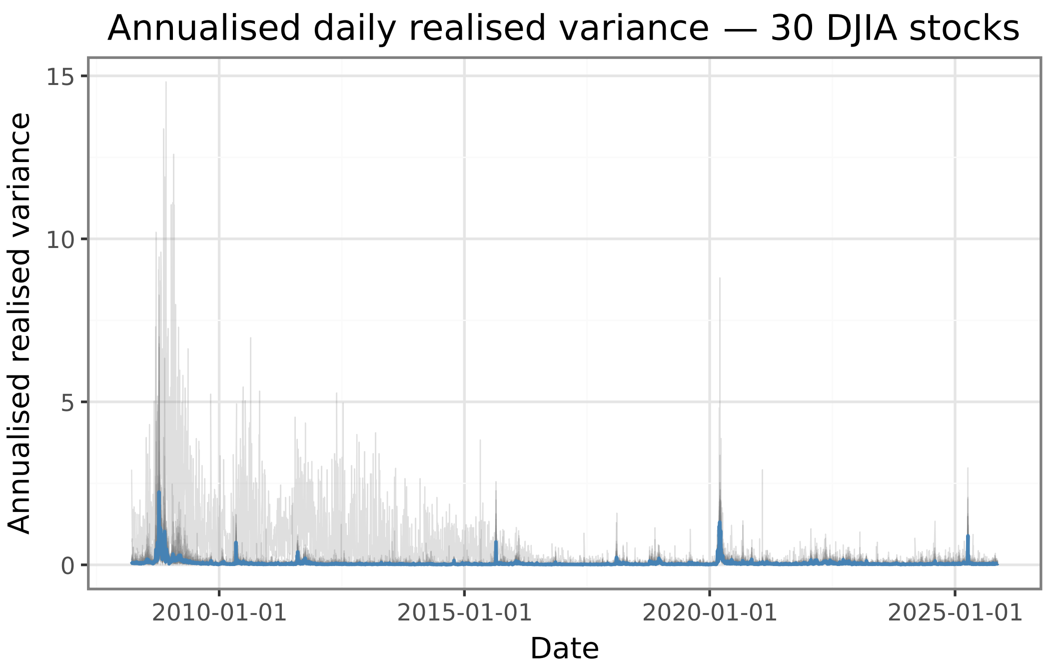

Figure 2: Annualised daily realised variance (diagonal of RC_d) for all 30 DJIA stocks (grey lines), cross-sectional median (blue) and IQR (band). Spikes mark the 2008–09 crisis, March 2020, and 2022.

The diagonal elements of RC_d (Figure 2) show three key patterns: (i) **volatility clustering** — large variance days cluster together; (ii) **positive skewness with slow decay** — a signature of the leverage effect; and (iii) **persistent cross-sectional heterogeneity** — some stocks carry materially more variance than others throughout the sample. The three most prominent spikes correspond to the 2008–09 global financial crisis, the March 2020 COVID shock, and the 2022 rate-hike episode. At one-minute sampling, off-diagonal elements would be biased toward zero by the **Epps effect** (Barndorff-Nielsen et al., 2011) — asynchronous trading attenuates cross-correlations — an issue that 5-minute sampling largely eliminates.

3 Exercise 3: Covariance Estimators

The evaluation window covers the second half of the full sample, running from January 17, 2017 to November 14, 2025.

(a) HF random walk. $\hat{\Sigma}_t^{HF} = RC_{t-1}$ assumes the covariance follows a **martingale** — no mean reversion, no drift — which is a natural baseline when intraday data are available. **(b) LW on daily returns.** We use $W = 252$ trading days, equivalent to one calendar year of trading. This window length reflects a bias–variance tradeoff: shorter windows reduce staleness when the true covariance shifts but increase estimation variance, while longer windows smooth out noise at the cost of delayed adaptation. One year gives $T/N \approx 8.4$, providing a full-rank sample covariance matrix while remaining timely enough to track gradual structural changes; Ledoit and Wolf (2003) linear shrinkage toward a scaled identity then further stabilises the precision matrix for portfolio inversion.

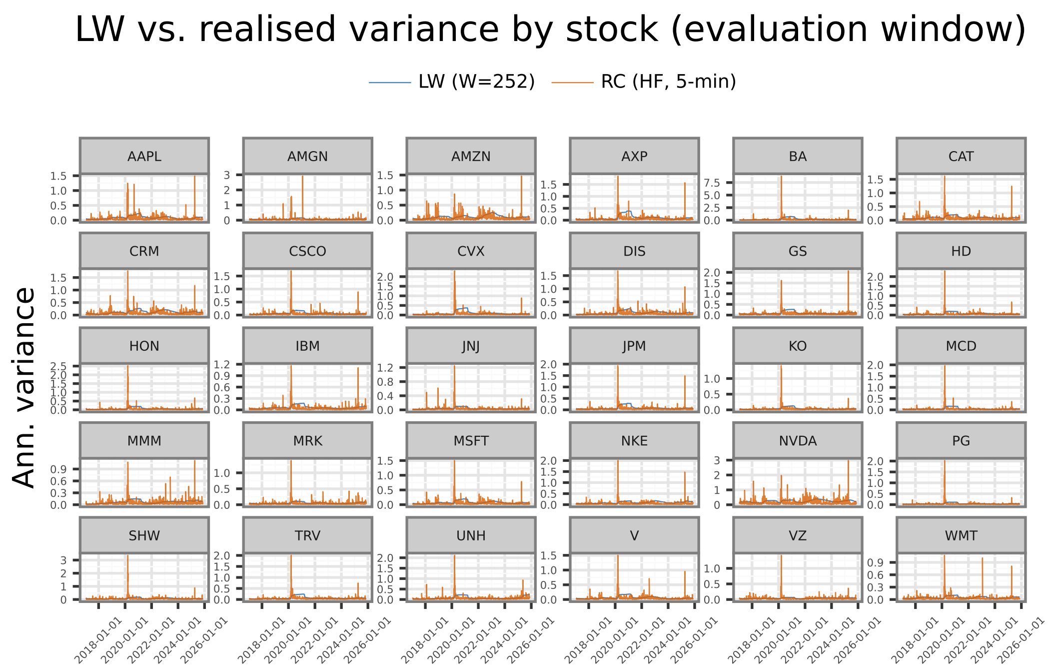

Figure 3: Per-stock annualised variance over the evaluation window for all 30 DJIA stocks. Each panel shows the LW-implied variance (orange, rolling 252-day Ledoit–Wolf shrinkage) and the realised variance from the diagonal of RC_t (blue, 5-min). LW is consistently smoother; RC reacts immediately to volatility bursts.

Figure 3 confirms the smoothing property of LW across all 30 stocks: the orange (LW) line is consistently flatter than the blue (RC) line, which spikes sharply during stress events such as March 2020 and 2022. The free y-axis scale reveals substantial cross-sectional heterogeneity — high-volatility stocks (e.g., NVDA, BA) show far larger absolute spikes than defensive names (e.g., KO, WMT) — a pattern that a single median would mask.

4 Exercise 4: Minimum-Variance Portfolio Weights

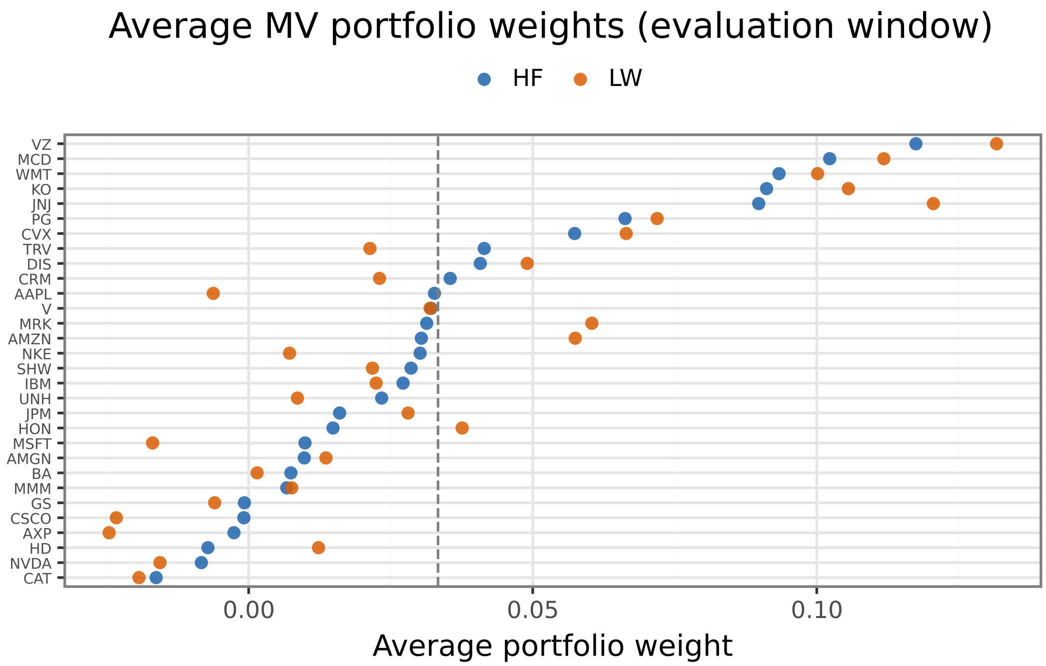

Figure 4: Time-averaged MV portfolio weights for the 30 DJIA stocks, ordered by HF average weight. Dashed line: equal weight (1/30).

The HF strategy assigns its largest average weight to **VZ** (0.117) and its most negative to **CAT** (-0.016). The LW strategy is most positive on **VZ** (0.132) and most negative on **AXP** (-0.025). Average daily turnover is **3.476** (HF) and **0.071** (LW), with the HF strategy showing higher turnover.

Both strategies hold extreme short and long positions — a known consequence of unconstrained minimum-variance optimisation where small errors in $\hat{\Sigma}$ cause large weight oscillations (Ledoit and Wolf, 2003). The HF strategy's higher turnover follows directly from the greater day-to-day variation in RC_{t-1} relative to the slow-moving LW estimate.

5 Exercise 5: Out-of-Sample Portfolio Evaluation

Table 1: Out-of-sample performance (second half of sample). Ann. volatility = s.d. $\times\sqrt{252}$; Sharpe ratio = ann. mean / ann. s.d. (risk-free rate = 0, no transaction costs).

Out-of-sample portfolio performance		
Strategy	Ann. Volatility	Ann. Sharpe
HF (RC, t-1)	0.167	0.466
LW (W = 252)	0.141	0.674
Equal-weighted (1/N)	0.179	0.763

Equal-weighted achieves the highest annualised Sharpe ratio (0.763); **HF (RC, t-1)** produces the lowest (0.466). The HF estimator does not outperform LW on Sharpe ratio (0.466 vs. 0.674).

Although each daily RC_d matrix is estimated from $K \approx 78$ five-minute intraday returns — well above the $N = 30$ stocks needed for full rank — the **random-walk estimator uses only a single day of data** per forecast. The LW estimator, by contrast, averages over $W = 252$ daily returns, smoothing out day-to-day noise at the cost of slower adaptation to genuine covariance shifts. In the relatively calm post-2016 evaluation window, this smoothing advantage of LW dominates the timeliness advantage of RC_{t-1}, so the LW estimate — while slower to react — produces more stable weights and, here, a higher Sharpe ratio.

The in-class discussion on realised kernels and portfolio choice suggests that HF covariance information translates into economically meaningful portfolio gains when (i) the universe is large ($N \gg 30$) so that estimation error in $\hat{\Sigma}$ binds, (ii) the true covariance changes rapidly, and (iii) information gains exceed turnover costs. With only 30 DJIA stocks, LW shrinkage of daily returns already captures the dominant covariance structure, leaving limited room for intraday data to add value (Ledoit and Wolf, 2004).

6 References

Andersen, T. G. and Bollerslev, T. (1998). Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. *International Economic Review*, 39(4):885–905.

Barndorff-Nielsen, O. E., Hansen, P. R., Lunde, A., and Shephard, N. (2011). Multivariate realised kernels: Consistent positive semi-definite estimators of the covariation of equity prices with noise and non-synchronous trading. *Journal of Econometrics*, 162(2):149–169.

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Ledoit, O. and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411.